

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION

SECTION II – Sanitary Wastewater

D. Nondomestic Wastewater Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257
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1. SEPTAGE – Does this facility accept septage?

- ☐ Yes. On a separate sheet, describe the allocation of the Maximum Allowable Headworks Loading (MAHL) capacity to domestic wastewater, nondomestic wastewater, and septage. The MAHL should include the treatment plant's design and current loading and, at a minimum, the number of gallons and concentrations of the pollutants BOD, TSS, PO₄, and NH₃ that are attributable to each wastewater.
- ☒ No. Continue with Item 2.

2. RESOURCE CONSERVATION AND RECOVERY ACT (RCRA) WASTEWATER

Does this facility receive, or has it in the last three (3) years received, RCRA hazardous waste by truck, rail, or dedicated pipe?

- ☐ Yes. Provide the following information on a separate sheet: The method by which the waste is received (e.g., truck, rail, or dedicated pipe), the waste's "EPA Hazardous Waste Number," and the amount of waste received in either mass or volume.
- ☒ No. Continue with Item 3.

3. REMEDIATION WASTEWATER

Does this facility receive, or has it been notified that it will receive in the next five (5) years, wastes from remedial activities?

- ☐ Yes. Provide a list on a separate sheet that contains the following information for each current and future remediation site:
- 1) Describe the site and type of facility at which the CERCLA/RCRA or other remedial waste originates or will originate.
 - 2) List the hazardous constituents that are, or are expected to be, received at the POTW. Include data on volume and concentration, if known.
 - 3) Describe in detail any treatment the waste receives before being discharged to the POTW.
 - 4) Provide the schedule for when the remediation wastewater is discharged to the POTW.
- ☒ No. Continue with Item 4.

4. INDUSTRIAL AND COMMERCIAL SOURCES

A. Does this facility receive any nondomestic wastewater from any industrial or commercial facilities? (Nondomestic wastewater refers to water that carries wastes other than human and household wastes.)

- ☒ Yes. Continue with Item B.
- ☐ No. Go to Part E. Biosolids Information.

B. Provide the following information:

- 1) Estimate the average volume of nondomestic wastewater received by this facility: 820,000 MGD
- 2) Describe the type of nondomestic wastewater(s) received by this facility in the space provided below.

Wastewater Type	Volume (MGD)	Wastewater Type	Volume (MGD)
☒ Industrial Process Wastewater	<u>210,000</u>	☒ Landfill Leachate	<u>33,600</u>
☒ Contact Cooling Water	<u>288,000</u>	☐ Trucked Industrial Wastewater	_____
☒ Noncontact Cooling Water	<u>288,000</u>	☐ Other: _____	_____

C. Is an Industrial Pretreatment Program (IPP) currently required by the DEQ? Note: Applicants with an IPP are required to also complete Item 5 on Page 15.

- ☒ Yes. Provide the most recent approval date for the following elements of the program:

Sewer Use Ordinance	<u>2001</u>	Interjurisdictional Agreements	<u>1968</u>	Procedures	<u>1995</u>
Other Legal Authority	<u>1995</u>	Enforcement Response Plan	<u>1995</u>	Local Limits	<u>1995</u>

- ☐ No.

Michigan Department of Environmental Quality – Water Resources Division
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E. Biosolids Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257
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1. **BIOSOLIDS HANDLING** – All facilities that generate or propose to generate biosolids must complete Items 1. and 2.
 Provide total English dry tons per 365-day period of residuals handled under the following practices:

Amount generated at the facility: <u>291.5</u>	Amount sent to municipal solid waste landfill: <u>0</u>
Amount received from off-site: <u>0</u>	Amount sold or given away in a bag or other container for application to the land: <u>0</u>
Amount treated on-site (including blending): <u>291.5</u>	Amount transported to another POTW: <u>0</u>
Amount used or disposed of by another practice: <u>0</u>	Transport Company: <u>0</u>
Amount applied to land in bulk form: <u>291.5</u>	Receiving POTW: <u>0</u>
Amount fired in incinerator: <u>0</u>	

BIOSOLIDS STORAGE
 Enter the volume of residual storage capacity at this facility: 1.5 ☒ million gallons or ☐ cubic feet

2. **LAND APPLICATION** – Facilities that land apply must complete Items A. - D., or have submitted a Biosolids Annual Report as required in the facility's current Residual Monitoring Program. Latest Biosolids Annual Report submitted on October 28, 2011

A. **BIOSOLIDS CHARACTERISTICS** – New Land Appliers Only
 Report one year of residuals monitoring data and in no case less than three (3) sampling events for the following parameters. Provide the actual analytical data sheets as an attachment. Analytical methods shall be in accordance with Rule 323.2406 (2) "Methods for Biosolids."

Parameter	Average Monthly Concentration	Maximum Concentration	Units	Number of Analyses	Sample Type	Analytical Method	Quantification Level
Total Solids	3.72	8.10	%	14	☐ Grab ☒ Composite	EPA 160.3	1 µg/l
Total Arsenic	29.8	30.9	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	1 µg/l
Total Cadmium	9.47	12.20	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	0.2 µg/l
Total Copper	1116.3	1470.0	mg/kg	3	☐ Grab ☒ Composite	EPA 6010B	1 µg/l
Total Lead	<115.13	<218	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	1 µg/l
Total Mercury	2.54	5.13	mg/kg	3	☐ Grab ☒ Composite	EPA 7471A	0.5 µg/l
Total Molybdenum	43.3	49.8	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	
Total Nickel	<34.5	41.3	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	5 µg/l
Total Selenium	<10.3	<20	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	1.0 µg/l
Total Zinc	1890	2850	mg/kg	5	☐ Grab ☒ Composite	EPA 6020	10 µg/l
Total Kjeldahl Nitrogen	128933	232000	mg/kg	3	☐ Grab ☒ Composite	EPA 351.2	
Ammonium Nitrogen	69000	146000	mg/kg	3	☐ Grab ☒ Composite	EPA 350.1	
Total Phosphorus	38600	51900	mg/kg	3	☐ Grab ☒ Composite	SM4500-PB5,E	
Total Potassium	6487	14100	mg/kg	3	☐ Grab ☒ Composite	EPA 6010B	

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B. POLLUTANTS OF CONCERN

Are there currently, or is there potential for, pollutants (other than the parameters listed on the previous page) to be present in the residuals at concentrations that would make them unsuitable for land application?

☐ Yes. On a separate sheet, provide representative analytical data for those pollutants.

☒ No. Continue with Item C.

C. ADDITIONAL BIOSOLIDS MONITORING DATA

Report any biosolids monitoring data from the last permit cycle for parameters not specifically listed on the previous page. Include the actual analytical data sheets as an attachment. Upon submittal review, additional monitoring may be required if the Water Resources Division has reason(s) to suspect that the information provided (or not provided) does not adequately characterize the residuals proposed to be land applied. For assistance with completing this item, contact the Permits Section. To submit additional information, see Page ii, Item 3.

Parameter	Average Monthly Concentration	Maximum Concentration	Units	Number of Analyses	Sample Type	Analytical Method	Quantification Level
ALUMINIUM	6250	8680	mg/kg	3	☐ Grab ☒ Composite	EPA 6010B	
CALCIUM	42367	78100	mg/kg	3	☐ Grab ☒ Composite	EPA 6010B	
CHROMIUM	123	168	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	
MAGNESIUM	6293	11900	mg/kg	3	☐ Grab ☒ Composite	EPA 6010B	
SILVER	10	11.7	mg/kg	3	☐ Grab ☒ Composite	EPA 6020	0.5 µg/l
SODIUM	11027	23800	mg/kg	3	☐ Grab ☒ Composite	EPA 6010B	
CHLORIDE	16383	33600	mg/kg	3	☐ Grab ☒ Composite	EPA 300.0A	
SULFATE	<1624	2490	mg/kg	3	☐ Grab ☒ Composite	EPA 300.0A	

D. LAND APPLICATION SITE LIST

Provide the following information for every new or existing site that may be used in the next five years (biosolids permit cycle). Each listed site should have been submitted to the DEQ on a Site Identification Form (with attachments) since January 1, 1998, or the required information should be included with this form. Additional sites may be added to the Land Application Site List during the biosolids permit cycle by submitting a completed Site Identification Form with the appropriate attachments and waiting the required ten-day notification period. To submit additional information, see Page ii, Item 3.

Site Identification Number	Latitude	Longitude	Acres	Owner's Last Name	New Site	CPLR Site
RJ001	44° 8' 31.03	-85° 12.22"	40	Jackson	☒	☐
RJ002	44° 8' 40.65	-85° 12.13"	55	Jackson	☒	☐
LP001	N44 15.266	W85 15.308	55	Parker	☐	☐
JE001	N4417.241	W085 13.274	40	Eubank	☐	☐
WE001	N44 17.340	W085 13.173	40	Eubank	☐	☐
LH001	N44 16.911	W085 14.941	34	Harington	☐	☐

This completes Section II. Return the completed Application (Sections I, II, IV, and any attachments) to one of the addresses on Page ii of this Application. If assistance is needed to complete this Application, contact the Permits Section.

E. Biosolids Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP				NPDES PERMIT NUMBER MI0020257		
6. Land Application Site List Provide the following information for every new or existing site that may be used in the next five years (biosolids permit cycle). Each listed site should either have had a Site Identification Form with attachments submitted to the MDEQ since January 1, 1998, or that information should be included with this form. Additional sites may be added to the Land Application Site List during the biosolids permit cycle by submitting a completed Site Identification Form with the appropriate attachments and waiting the required ten day notification period. Use additional pages as necessary.						
Site Identification Number	Latitude (dd,mm,ss)	Longitude (dd,mm,ss)	Acres	Owners Last Name	New Site? (check if yes)	CPLR* Site? (check if yes)
LH002	N44 16.930	W085 17.205	18	HARRINGTON	☐	☐
LH003	N44 16.911	W085 13.607	16	HARRINGTON	☐	☐
LH004	N44 17.061	W085 13.591	6.5	HARRINGTON	☐	☐
PM001	N44 11.208	W085 20.372	7	MILLER	☐	☐
IV01	N44 14.567	W085 17.589	33	VANDERVLUGT	☐	☐
IV02	N44 14.429	W085 17.414	18	VANDERVLUGT	☐	☐
IV03	N44 14.308	W085 17.589	37	VANDERVLUGT	☐	☐
IV04	N44 14.00	W085 16.900	16	VANDERVLUGT	☐	☐
IV05	N44 13.940	W085 16.900	16	VANDERVLUGT	☐	☐
IV06	N44 13.877	W085 16.900	16	VANDERVLUGT	☐	☐
IV07	N44 13.811	W085 16.900	14	VANDERVLUGT	☐	☐
IV08	N44 13.735	W085 16.900	15	VANDERVLUGT	☐	☐
IV09	N44 14.030	W085 16.550	8	VANDERVLUGT	☐	☐
IV10	N44 13.940	W085 16.471	7	VANDERVLUGT	☐	☐
IV11	N44 14.429	W085 17.360	12	VANDERVLUGT	☐	☐
IV12	N44 14.429	W085 17.300	6	VANDERVLUGT	☐	☐
IV13	N44 14.429	W085 14.240	6	VANDERVLUGT	☐	☐
IV14	N44 14.429	W085 17.625	8	VANDERVLUGT	☐	☐
IV15	N44 14.279	W085 17.035	6	VANDERVLUGT	☐	☐
KL01	N44 14.279	W085 17.035	46	LIEDEKE	☐	☐
CB01	N44 14.279	W085 15.649	25	BALDWIN	☐	☐
CB02	N44 15.696	W085 15.579	23	BALDWIN	☐	☐

* see definitions

KAR Laboratories, Inc.



4425 Manchester Road
Kalamazoo, MI 49001
Phone 269 381-9666
Fax 269 381-9698
www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

KAR Project No. : 111792
Date Reported : 05/10/11
Date Activated : 05/06/11
Date Due : 05/10/11
Date Validated : 05/10/11

Attn : Ms. Amy Vail

Project

Description : Analysis of one sludge sample for Spring Biosolids.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 111792. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,


David R. Alkema
Laboratory Manager

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KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **111792**

Date Reported : **05/10/11**

Attest:


David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample for Spring Biosolids.*

Sample ID : **"Sig-1"**

Sampled By : *HR of City of Cadillac, Water Resources*

Date Received : **05/06/11**

Sample Date : **05/05/11**

Sample Type : **sludge**

Sample Time : **1341**

KAR Sample No. : **111792-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	05/10/11	PML	
Prep, metals	Completed		EPA 3050	05/06/11	PML	
Aluminum, total	2900	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Arsenic, total	30.1	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Cadmium, total	8.7	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 0.6
Calcium, total	16,500	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Chromium, total	108	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Copper, total	976	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 30
Lead, total, by ICP	35.4	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Magnesium, total	2290	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Mercury, total	1.66	mg/kg total solids	EPA 7471A	05/10/11	PML	Detection limit = 0.1
Molybdenum, total	43.0	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Nickel, total	25.3	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Potassium, total	1800	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Selenium, total	6.0	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Silver, total	7.2	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 3
Sodium, total	2900	mg/kg total solids	EPA 6010B	05/09/11	PML	Detection limit = 100
Zinc, total	1710	mg/kg total solids	EPA 6020	05/09/11	PML	Detection limit = 30
Chloride	6700	mg/kg total solids	EPA 300.0A	05/06/11	ALK	Detection limit = 138
Nitrogen, ammonia	28,700	mg/kg total solids	EPA 350.1	05/10/11	ALK	Detection limit = 689
Nitrogen, nitrate	<28	mg/kg total solids	EPA 353.2	05/06/11	ALK	
Nitrogen, total kjeldahl	80,900	mg/kg total solids	EPA 351.2	05/06/11	JHB	Detection limit = 2760
Phosphorus, total (as P)	37,800	mg/kg total solids	SM 4500-P B5,E	05/06/11	JHB	Detection limit = 1380
Solids, total	3.63	% by weight	EPA 160.3	05/06/11	JHB	Detection limit = 0.01
Solids, volatile	59.55	% of total solids	EPA 160.4	05/06/11	JHB	Detection limit = 0.01
Sulfate	1920	mg/kg total solids	EPA 300.0A	05/06/11	ALK	Detection limit = 276
Waste pH	7.9	S.U.	EPA 9045C	05/06/11	HES	

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Phone 269 381-9666

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

KAR Project No. : 111573
Date Reported : 05/04/11
Date Activated : 04/20/11
Date Due : 05/04/11
Date Validated : 05/04/11

Attn : Ms. Amy Vail

Project

Description : Analysis of one sludge sample for Spring Biosolids.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 111573. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'David R. Alkema'.

David R. Alkema
Laboratory Manager

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KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **111573**

Date Reported : **05/04/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample for Spring Biosolids.*

Sample ID : "Tank #4"						
Sampled By : <i>DG of City of Cadillac, Water Resources</i>				Date Received : 04/20/11		
Sample Date : 04/18/11				Sample Type : sludge		
Sample Time :				KAR Sample No. : 111573-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	04/27/11	DBL	
Prep, metals	Completed		EPA 3050	04/26/11	PML	
Aluminum, total	8680	mg/kg total solids	EPA 6010B	04/29/11	DBL	Detection limit = 434
Arsenic, total	30.9	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 20
Cadmium, total	12.2	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 4
Calcium, total	78,100	mg/kg total solids	EPA 6010B	04/29/11	DBL	Detection limit = 434
Chromium, total	168	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 20
Copper, total	1470	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 20
Lead, total, by ICP	<218	mg/kg total solids	EPA 6010B	04/29/11	DBL	
Magnesium, total	11,900	mg/kg total solids	EPA 6010B	04/29/11	DBL	Detection limit = 434
Mercury, total	5.13	mg/kg total solids	EPA 7471A	04/27/11	DBL	Detection limit = 0.28
Molybdenum, total	49.8	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 20
Nickel, total	41.3	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 20
Potassium, total	14,100	mg/kg total solids	EPA 6010B	04/29/11	DBL	Detection limit = 434
Selenium, total	<20	mg/kg total solids	EPA 6020	04/28/11	PML	Elevated detection limit due to low solids content.
Silver, total	11.7	mg/kg total solids	EPA 6020	04/28/11	PML	Detection limit = 4
Sodium, total	23,800	mg/kg total solids	EPA 6010B	04/29/11	DBL	Detection limit = 4340
Zinc, total	2850	mg/kg total solids	EPA 6010B	04/29/11	DBL	Detection limit = 87
Chloride	33,600	mg/kg total solids	EPA 300.0A	04/29/11	ALK	Detection limit = 893
Nitrogen, ammonia	146,000	mg/kg total solids	EPA 350.1	05/03/11	ALK	Detection limit = 17,900
Nitrogen, nitrate	<18	mg/kg total solids	EPA 353.2	04/22/11	DRA	
Nitrogen, total kjeldahl	232,000	mg/kg total solids	EPA 351.2	05/02/11	JHB	Detection limit = 8930
Phosphorus, total (as P)	51,900	mg/kg total solids	SM 4500-P B5,E	04/28/11	JHB	Detection limit = 1790
Solids, total	0.56	% by weight	EPA 160.3	04/21/11	JHB	Detection limit = 0.01
Solids, volatile	58.44	% of total solids	EPA 160.4	04/21/11	JHB	Detection limit = 0.01
Sulfate	2490	mg/kg total solids	EPA 300.0A	04/29/11	ALK	Detection limit = 1790
Waste pH	8.1	S.U.	EPA 9045C	05/04/11	GJE	

Resampled this set of data because of low solids Percentage, suspected erroneous results, higher than our previous recorded results.



KAR Laboratories, Inc.

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Kalamazoo, MI 49001

Phone 269 381-9666

Fax 269 381-9698

www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : Mr. Jeff Dietlin

KAR Project No. : 104002

Date Reported : 10/08/10

Date Activated : 09/24/10

Date Due : 10/08/10

Date Validated : 10/08/10

Project

Description : Analysis of one sludge sample for Fall Biosolids.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 104002. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink that reads 'David R. Alkema'.

David R. Alkema
Laboratory Manager

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KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **104002**

Date Reported : **10/08/10**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample for Fall Biosolids.*

Sample ID : "Biosolids"						
Sampled By : <i>DG of City of Cadillac, Water Resources</i>				Date Received : 09/24/10		
Sample Date : 09/23/10				Sample Type : sludge		
Sample Time : 1315				KAR Sample No. : 104002-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	10/06/10	DBL	
Prep, metals	Completed		EPA 3050	09/29/10	PML	
Aluminum, total	7170	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 185
Arsenic, total	28.5	mg/kg total solids	EPA 6020	09/30/10	PML	Detection limit = 6
Cadmium, total	7.5	mg/kg total solids	EPA 6020	09/30/10	PML	Detection limit = 1
Calcium, total	32,500	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 185
Chromium, total	93	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 19
Copper, total	903	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 37
Lead, total, by ICP	<92	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 92
Magnesium, total	4690	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 185
Mercury, total	0.84	mg/kg total solids	EPA 7471A	10/06/10	DBL	Detection limit = 0.15
Molybdenum, total	37	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 19
Nickel, total	<37	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 37
Potassium, total	3560	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 185
Selenium, total	<5	mg/kg total solids	EPA 6020	09/30/10	PML	Detection limit = 5
Silver, total	11.1	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 9.3
Sodium, total	6380	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 1850
Zinc, total	1660	mg/kg total solids	EPA 6010B	10/06/10	DBL	Detection limit = 37
Chloride	8850	mg/kg total solids	EPA 300.0A	10/02/10	DMC	Detection limit = 231
Nitrogen, ammonia	32,300	mg/kg total solids	EPA 350.1	10/07/10	DMC	Detection limit = 926
Nitrogen, nitrate	<4.7	mg/kg total solids	EPA 353.2	09/24/10	ALK	Detection limit = 4.7
Nitrogen, total kjeldahl	73,900	mg/kg total solids	EPA 351.2	09/28/10	JHB	Detection limit = 2320
Phosphorus, total (as P)	26,100	mg/kg total solids	SM 4500-P B5,E	10/05/10	JHB	Detection limit = 470
Solids, total	2.16	% by weight	EPA 160.3	09/27/10	JHB	Detection limit = 0.01
Solids, volatile	56.16	% of total solids	EPA 160.4	09/27/10	JHB	Detection limit = 0.01
Sulfate	<463	mg/kg total solids	EPA 300.0A	10/02/10	DMC	Elevated detection limit due to sample matrix interference. Detection limit = 463
Waste pH	8.2	S.U.	EPA 9045C	10/06/10	HES	

KAR Laboratories, Inc.

KAR Laboratories, Inc.



4425 Manchester Road

Kalamazoo, MI 49001

Phone 269 381-9666

Fax 269 381-9698

www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

KAR Project No. : 113351

Date Reported : 08/23/11

Date Activated : 08/09/11

Date Due : 08/23/11

Date Validated : 08/23/11

Attn : Ms. Amy Vail

Project

Description : Analysis of one sludge sample for Fall Biosolids & Monitoring.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 113351. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink that reads 'David R. Alkema'.

David R. Alkema
Laboratory Manager

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KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **113351**

Date Reported : **08/23/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample for Fall Biosolids & Monitoring.*

Sample ID : <u>"Digester #4"</u>						
Sampled By : <i>DL of City of Cadillac, Water Resources</i>				Date Received : 08/08/11		
Sample Date : 08/04/11				Sample Type : sludge		
Sample Time : 1300				KAR Sample No. : 113351-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, metals</i>	<i>Completed</i>		<i>EPA 3050</i>	<i>08/16/11</i>	<i>PML</i>	
<i>Zinc, total</i>	<i>1550</i>	<i>mg/kg total solids</i>	<i>EPA 6010B</i>	<i>08/19/11</i>	<i>DBL</i>	<i>Detection limit = 20</i>
<i>Solids, total</i>	<i>4.27</i>	<i>% by weight</i>	<i>EPA 160.3</i>	<i>08/10/11</i>	<i>JWW</i>	<i>Detection limit = 0.01</i>

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Phone 269 381-9666

Fax 269 381-9698

www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

KAR Project No. : 104242
Date Reported : 10/22/10
Date Activated : 10/11/10
Date Due : 10/25/10
Date Validated : 10/22/10

Attn : Ms. Amy Vall

Project

Description : Analysis of one sludge sample.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 104242. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink that reads 'David R. Alkema'.

David R. Alkema
Laboratory Manager

KAR Laboratories, Inc. maintains Full Certification status for Bacteriology, Inorganics, Regulated Organics and Synthetic Organics through USEPA, Michigan Department of Public Health and Indiana State Department of Health. This report may only be reproduced in full and not without the written consent of City of Cadillac, Water Resources.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **104242**

Date Reported : **10/22/10**

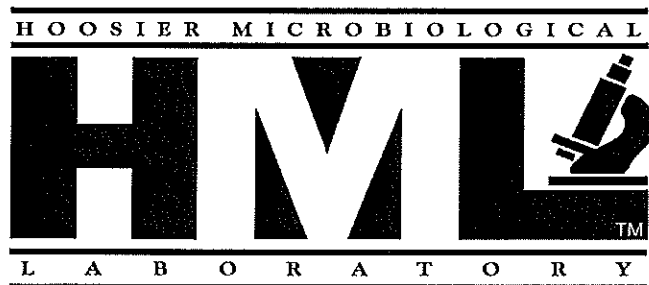
Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample.*

Sample ID : <u>"Biosolids"</u>						
Sampled By : <i>RM of City of Cadillac, Water Resources</i>				Date Received : 10/11/10		
Sample Date : 10/07/10				Sample Type : sludge		
Sample Time : 1500				KAR Sample No. : 104242-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, metals</i>	<i>Completed</i>		<i>EPA 3050</i>	<i>10/13/10</i>	<i>PML</i>	
<i>Zinc, total</i>	<i>1680</i>	<i>mg/kg total solids</i>	<i>EPA 6010B</i>	<i>10/20/10</i>	<i>DBL</i>	<i>Detection limit = 39</i>
<i>Solids, total</i>	<i>2.25</i>	<i>% by weight</i>	<i>EPA 160.3</i>	<i>10/13/10</i>	<i>JHB</i>	<i>Detection limit = 0.01</i>

KAR Laboratories, Inc.



Testing • Research • Consulting

Multiple Analysis Report

Sample: 226293

November 2, 2010

Ms. Amy Vail
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Biosolids
1121 Plett Rd.
Cadillac, MI 49601

Dear Ms. Vail:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 9:23 AM, 09/24/2010, and collected at 1:05 PM, 09/23/2010:

<u>TEST - METHOD</u>	<u>RESULT</u>	<u>MDL*</u>	<u>Date Complete</u>
Fecal Coliform-SM9221E (MPN)	790 MPN/g dry	1 MPN/g dry	09/28/2010
Enteric Virus-EPA 625/R-92/013	<1 pfu/4g dry	1 pfu/4g dry	10/28/2010
Solids, total-SM 2540G	2.15%	0.005%	10/21/2010
Salmonella-EPA 625/R-92/013	<3 MPN/4 g dry	3 MPN/4 g dry	09/27/2010
Helminth Ova-EPA/625/R-92/013	<1 ovum/4g dry	1 ovum/4 g dry	10/18/2010

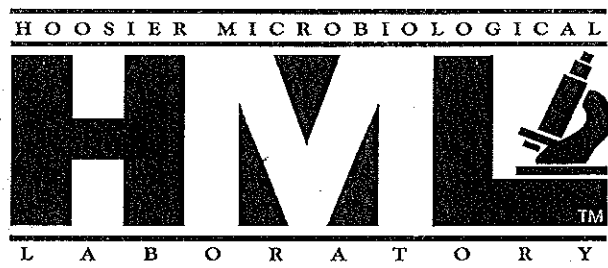
*Minimum Detection Level

This testing was completed by D.S. and M.E. Please feel free to contact us if we can be of further service to you.

Sincerely,

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp



Testing • Research • Consulting

Multiple Analysis Report

Sample: 234812

October 27, 2011

Ms. Amy Vail
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Tank 4
1121 Plett Rd.
Cadillac, MI 49601

Dear Ms. Vail:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 10:29 AM, 4/19/2011, and collected at 1:00 PM, 4/18/2011:

<u>TEST - METHOD</u>	<u>RESULT</u>	<u>MDL*</u>	<u>Date Complete</u>
Fecal Coliform-SM9221E (MPN)	550 MPN/g dry	1 MPN/g dry	4/22/2011
Enteric Virus-EPA 625/R-92/013	<1 pfu/4g dry	1 pfu/4g dry	5/20/2011
Solids, total-SM 2540G	0.60%	0.005%	4/22/2011
Salmonella-EPA 625/R-92/013	<3 MPN/4 g dry	3 MPN/4 g dry	4/21/2011
Helminth Ova-EPA/625/R-92/013	<1 ovum/4g dry	1 ovum/4 g dry	5/16/2011

*Minimum Detection Level

This testing was completed by D.S. and A.G. Please feel free to contact us if we can be of further service to you.

Sincerely,

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp